ex 1 E.M. in Minkowski space

$\vec{E}$ electric field

$\vec{B}$ magnetic field

Maxwell's eq:

$$\begin{cases} \nabla \cdot E = \rho \\ \nabla \cdot B = 0 \quad \checkmark \\ \nabla \times E = -\frac{\partial B}{\partial t} \quad \checkmark \\ \nabla \times B = J + \frac{\partial E}{\partial t} \end{cases}$$

potentials

in $\mathbb{R}^4$ $\begin{cases} \text{if } \nabla \cdot F = 0 \text{ then } F = \nabla \times A \text{ for some } A \to \text{vector} \\ \text{if } \nabla \times F = 0 \text{ then } F = \nabla \varphi \text{ for some } \varphi \to \text{scalar} \end{cases}$

$\nabla \cdot B = 0 \rightarrow B = \nabla \times A$

$\nabla \times E = -\frac{\partial}{\partial t}(\nabla \times A) \rightarrow \nabla \times \left(E + \frac{\partial A}{\partial t}\right) = 0 \rightarrow E = -\nabla \varphi - \frac{\partial A}{\partial t}$

$$\nabla \cdot E = \boxed{-\nabla^2 \varphi - \frac{\partial}{\partial t} \nabla \cdot A = \rho}$$

$$\nabla \times B = \boxed{\nabla \times \nabla \times A = J - \nabla \frac{\partial \varphi}{\partial t} - \frac{\partial^2 A}{\partial t^2}}$$

gauge invariance: if $\begin{cases} A \to A + \nabla \lambda \\ \varphi \to \varphi - \frac{\partial \lambda}{\partial t} \end{cases}$ for some scalar field λ

then E and B are the same

$$\nabla \times (A + \nabla \lambda) = \nabla \times A + \underbrace{\nabla \times \nabla \lambda}_{\to 0} = \nabla \times A$$

$$-\nabla\left(\varphi - \cancel{\frac{\partial \lambda}{\partial t}}\right) - \frac{\partial}{\partial t}\left(A + \cancel{\nabla \lambda}\right) = -\nabla \varphi - \frac{\partial A}{\partial t}$$

B pseudo-vector field $\rightarrow$ transforms "weird" under reflections

Differential forms

k-forms on M $n = \dim(M)$

Hodge star: note that $\Lambda^k(M)$ and $\Lambda^{n-k}(M)$ are isomorphic

dimension of $\Lambda^k(M) = \binom{n}{k}$

for example $\Lambda^0(M)$ is 1-d

if $\alpha \in \Lambda^n(M)$ then $\alpha \propto dx^1 \wedge dx^2 \wedge \dots \wedge dx^n$

Hodge star is a specific isomorphism that we can build once we have a metric on M

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on $\mathbb{R}^4$ t, x, y, z coordinates $(c=1)$

$* dt = -dx \wedge dy \wedge dz$

$* dx = -dt \wedge dy \wedge dz$

$* dy = -dt \wedge dz \wedge dx$

$* dz = -dt \wedge dx \wedge dy$

$*(*\alpha) = -(-1)^{|\alpha|} \alpha$

$* 1 = dt \wedge dx \wedge dy \wedge dz$

$*(dt \wedge dx) = -dy \wedge dz$

$*(dt \wedge dy) = -dz \wedge dx$

$*(dt \wedge dz) = -dx \wedge dy$

$*(dx \wedge dy) = dt \wedge dy$

$*(dy \wedge dz) = dt \wedge dx$

$*(dz \wedge dx) = dt \wedge dy$

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if $\alpha = \alpha_x dx + \alpha_y dy + \alpha_z dz$

$$d\alpha = \partial_t \alpha_x \, dt\wedge dx + \partial_t \alpha_y \, dt\wedge dy + \partial_t \alpha_z \, dt\wedge dz$$

$$(\partial_x \alpha_y - \partial_y \alpha_x) dx\wedge dy + (\partial_y \alpha_z - \partial_z \alpha_y) dy\wedge dz + (\partial_z \alpha_x - \partial_x \alpha_z) dz\wedge dx$$

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let $\quad E = E_x dx + E_y dy + E_z dz$

$$B = B_x \, dy\wedge dz + B_y \, dz\wedge dx + B_z \, dx\wedge dy$$

<u>put them together</u> $\quad F = B + E\wedge dt \quad$ $\boxed{\text{Faraday tensor}}$

$$dF = dB + dE\wedge dt$$

$$dB = \partial_t B_x \, dt\wedge dy\wedge dz + \partial_t B_y \, dt\wedge dz\wedge dx + \partial_t B_z \, dt\wedge dx\wedge dy$$

$$\nabla\cdot\vec{B} \; dx\wedge dy\wedge dz$$

$$\nabla\cdot E = 0$$

$$\nabla\cdot B = 0 \quad ✓$$

$$\nabla\times E = -\frac{\partial B}{\partial t} \quad ✓$$

$$\nabla\times B = \frac{\partial E}{\partial t}$$

$$dE\wedge dt =$$

$$\underbrace{(\partial_x E_y - \partial_y E_x)}_{(\nabla\times E)_z} dt\wedge dx\wedge dy + \underbrace{(\partial_y E_z - \partial_z E_y)}_{(\nabla\times E)_x} dt\wedge dy\wedge dz + \underbrace{(\partial_z E_x - \partial_x E_z)}_{(\nabla\times E)_y} dt\wedge dz\wedge dx$$

$$dF = 0 \rightarrow \begin{cases} \nabla\cdot\vec{B} = 0 \\ \nabla\times\vec{E} + \dfrac{\partial\vec{B}}{\partial t} = 0 \end{cases}$$

$$(\nabla\times F)_i = \sum_{jk} \varepsilon_{ijk} \partial_j F_k$$

closed $\Rightarrow$ exact

if $dF = 0$ then $F = dA$ $\quad A$ is a 1-form

curvature of connection $\qquad$ connection in gauge theory

$$A = \varphi \, dt + A_x dx + A_y dy + A_z dz$$

from $\vec{B} = \nabla\times\vec{A}$ we have $\vec{A} = (A_x, A_y, A_z)$

$\vec{E} = -\nabla\varphi - \frac{\partial A}{\partial t}$ we have $A_t = \varphi$

$$dA = (\partial_t A_x - \partial_x \varphi)\, dt \wedge dx + (\partial_t A_y - \partial_y \varphi)\, dt \wedge dy + (\partial_t A_z - \partial_z \varphi)\, dt \wedge dz$$
$$+ (\nabla \times \vec{A})_z\, dx \wedge dy + (\nabla \times \vec{A})_y\, dz \wedge dx + (\nabla \times \vec{A})_x\, dy \wedge dz$$

$F = dA \rightarrow \quad B + E \wedge dt = dA \quad \Leftrightarrow \quad \vec{B} = \nabla \times \vec{A}$

$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}$

gauge invariance : $\quad A \rightarrow A + d\lambda \qquad F \rightarrow F$

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build principal $U(1)$-bundle out of Minkowski space and the bad gauge field

define $\quad A = i\omega_1 \qquad \omega_1$ is a $\underline{u}(1) \equiv i\mathbb{R}$ valued 1-form

if there is a function $t_{12} : \mathbb{R}^4 \rightarrow U(1)$ and we define

$$\omega_2 = t_{12}^{-1}\, \omega_1\, t_{12} - t_{12}^{-1}\, dt_{12} = \omega_1 - t_{12}^{-1}\, dt_{12}$$

then ω_2 describes the same connection

if $\tilde{A} = i\omega_2$, how are $\tilde{A}$ and A related?

suppose that $\quad t_{12} = e^{i\theta} \qquad \theta : \mathbb{R}^4 \rightarrow \mathbb{R}$

$$dt_{12} = i e^{i\theta}\, d\theta \qquad t_{12}^{-1}\, dt_{12} = i\, d\theta$$

$$\tilde{A} = i\omega_2 = i(\omega_1 - i\, d\theta) = A + d\theta$$